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# Airport Service Quality as an Integrated Infrastructure System: A Systematic Literature Review of Service Domains, Passenger Heterogeneity, Evaluation Frameworks, and Industry Benchmarking

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**Abstract:** Airport service quality (ASQ) plays a central role in shaping passenger satisfaction, airport competitiveness, and the long-term performance of aviation infrastructure systems. Existing studies increasingly recognise airports not merely as service environments but as integrated infrastructure systems in which transportation networks, operational processes, information systems, and passenger-facing functions interact dynamically. However, current knowledge remains fragmented across service domains, evaluation methods, passenger characteristics, and industry benchmarking practices. This study investigates the analytical foundations and research evolution of ASQ from an infrastructure systems perspective. A systematic literature review was conducted following the PRISMA protocol using a Scopus-based corpus of 303 peer-reviewed publications published between 1976 and 2024, supplemented by major industry reports and benchmarking frameworks. The review synthesised evidence across airport service domains, service attributes, and key performance indicators (KPIs), passenger heterogeneity, survey methodologies, social media-based assessment approaches, and the interaction between airline and ASQ. The results showed that ASQ research has evolved from isolated service evaluation toward increasingly integrated and multi-dimensional assessment frameworks. Processing and non-processing service domains were found to exert asymmetric effects on passenger satisfaction, while substantial variations were identified across demographic, behavioural, geographic, and travel-related passenger profiles. The review further showed that industry benchmarking systems provide operational comparability but remain only partially aligned with academic analytical approaches. Several research gaps were identified, particularly in landside infrastructure evaluation, arrival-stage service assessment, integrated objective–subjective performance measurement, and system-level understanding of airport operations. The findings indicate that ASQ should be interpreted as an emergent property of interconnected infrastructure subsystems rather than as isolated service encounters. This study provides an integrated conceptual foundation for future research on intelligent, resilient, and evidence-based airport infrastructure management and supports more transparent and analytically grounded decision-making for airport operators, policymakers, and researchers.

**Keywords:** Airport service quality; Infrastructure systems; Integrated infrastructure; Passenger satisfaction; Service quality assessment; Key performance indicators; Airport performance evaluation; Systematic literature review

## 1 Introduction

Airports are among the most complex service environments in the world, serving as multi-functional nodes that simultaneously deliver processing services (check-in, security, immigration, boarding) and non-processing commercial and comfort experiences (retail, food and beverage (F&B), lounges, wayfinding). The quality of these services profoundly shapes passenger satisfaction, loyalty, and ultimately the commercial and competitive performance of airports [1–3].

The study of airport service quality (ASQ) has grown substantially since the foundational work of Parasuraman et al. [4] on service quality and the subsequent adaptation of these frameworks to the airport context. Early studies focused primarily on terminal LOS using physical and operational metrics [2, 5, 6]. More recent work integrates subjective passenger perceptions, emotional dimensions of the travel experience, and digital feedback from social media platforms [7].

Despite this expanding body of research, several important questions remain underexplored. These include: How do different service domains contribute differently to overall satisfaction? How do passengers of different profiles—by age, gender, trip purpose, nationality, or travel class—evaluate the same service attributes differently? How do academic survey instruments compare with industry indices such as the airports council international (ACI) ASQ Programme? What role has social media played in reshaping how ASQ is measured? And critically, how does airline service quality interact with and influence perceptions of ASQ?

This paper addresses the above-cited questions systematically. Section 2 describes the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) based methodology common to both papers in this series. Section 3 provides an abbreviated bibliometric overview. Sections 4 through 12 present the detailed domain-level analysis, passenger profiling findings, survey methodology review, social media evidence, airline–airport interface, and comparative industry index assessment. Sections 13 and 14 discuss research gaps and conclusions.

Airports are complex, integrated infrastructure systems comprising interdependent transportation networks, operational processes, information systems, and passenger service functions. The performance and efficiency of an airport depend on the seamless interaction and coordination among these interconnected subsystems (Figure 1).

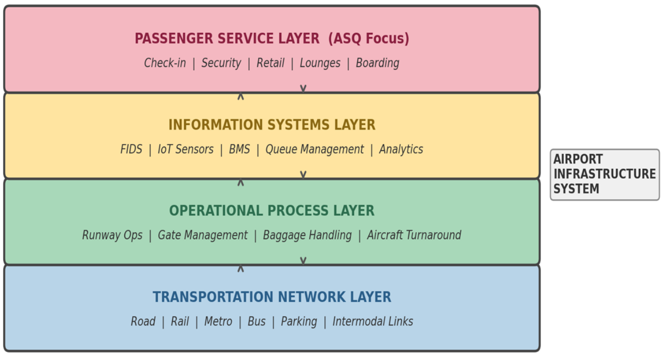


**Figure 1.** Airport as an integrated infrastructure system

Note: ASQ—airport service quality; FIDS—flight information display systems; IoT—Internet of Things; BMS—building management systems.

This systems perspective reveals that service quality outcomes are not solely determined by service design or staff performance, but are fundamentally shaped by infrastructure capacity, network connectivity, operational coordination, and information system maturity. A conceptual illustration of these layered interactions is presented below:

The information systems layer—encompassing flight information display systems (FIDS), passenger flow monitoring, queue management technologies, building management systems (BMS), and increasingly, Internet of Things (IoT)-enabled sensor networks—provides the digital backbone that enables real-time coordination across the physical infrastructure. Finally, the passenger service layer—the primary focus of ASQ research—operates within and is constrained by the capabilities of the underlying infrastructure layers.

At the transportation network layer, airports serve as intermodal nodes connecting air, road, rail, and public transit systems. The efficiency of landside access infrastructure—highway connections, metro/rail links, bus services, and parking facilities—directly shapes the pre-departure and post-arrival passenger experience. At the operational process layer, airside operations (runway allocation, gate management, baggage handling systems, aircraft turnaround) interact with terminal processing flows (check-in, security, immigration, boarding) through shared resources and timing constraints. Delays or disruptions at any node propagate through the system, affecting service quality across multiple domains simultaneously.

While the preceding discussion frames airports primarily as service environments, it is essential to recognise that airports function as complex, integrated infrastructure systems. An airport is not merely a collection of service touchpoints but a multi-layered system where transportation networks, operational processes, information systems, and passenger service functions interact dynamically. Understanding these interdependencies is critical for infrastructure-oriented research and practice.

## 2 Methodology of Systematic Literature Review

This review follows the **PRISMA** protocol. The search was conducted on Scopus using a refined Boolean search string targeting service quality, passenger satisfaction, and airport/aviation contexts. The search was limited to

English-language peer-reviewed articles and reviews in Business, Management, Social Sciences, Engineering, and Decision Sciences. Table 1 shows the paper selection criteria.

**Table 1.** Criteria for paper selection

| Parameter | Criteria |
|---|---|
| Database | Scopus |
| Search fields | Title, Abstract, Keywords |
| Search string | ("service qualit" Or "service excellence" Or "ASQ index" Or "performance") AND ("passenger satisfaction" Or "customer satisfaction") AND (airport Or "air travel" Or aviation) |
| Document types | Peer-reviewed journal articles |
| Language | English |
| Time span | 1976–2024 |
| Initial results | 505 documents |
| After type/language/exclusion filters | 456 documents |
| After quality assessment and relevance screening | 303 documents |

Note: ASQ—airport service quality.

The final corpus of 303 documents was supplemented by industry reports (ACI, IATA, ACRP) and foundational theoretical works in service quality and consumer satisfaction research [4, 8–13] that, while not meeting all inclusion criteria, provide the theoretical scaffolding for the field.

## 3 Bibliometric Overview

The corpus spans 48 years (1976–2024), comprising 284 original research articles and 19 review papers from 161 journals. Key bibliometric descriptors are summarised in Table 2. Publication output was negligible before 2000, rose steadily from 2001, and surged post-2015, reflecting the global growth of low-cost carriers (LCCs), airport privatisation, digital feedback platforms, and increased academic attention to service management in infrastructure sectors. The annual growth rate of 8.25% exceeds the general growth of social science publication output, indicating a genuinely expanding and prioritised field, as illustrated in Figure 2.

**Table 2.** Bibliometric descriptors of the ASQ research corpus

| Descriptor | Value | Interpretation |
|---|---|---|
| Total documents | 303 | Large, representative corpus |
| Time span | 1976–2024 | 48 years; accelerating since 2015 |
| Sources (journals) | 171 | Broad interdisciplinary spread |
| Annual growth rate | 8.25% | Above-average for the social sciences |
| Average citations per document | 22 | Moderate-high academic impact |
| Total authors | 777 | Diverse, collaborative field |
| Average co-authors per document | 2.81 | Predominantly 2–3 author teams |
| International co-authorship | 22.77% | Growing international collaboration |
| Leading journal | *J. Air Transport Management* (33 papers, 10.9%) | Clear disciplinary home journal |
| Top research country | China (132 papers) | Dominates through big data/social media studies |

Note: ASQ—airport service quality.

The dominance of the *Journal of Air Transport Management* (Elsevier) as the primary outlet for ASQ research, followed by Sustainability and Research in Transportation Business and Management, reflects the applied, management-oriented nature of the field. Notably, 73% of publications are spread across 153 other journals, indicating substantial interdisciplinary reach into engineering, computer science, and social psychology (Figure 3). The most prolific individual researchers are Eboli and Mazzulla (5 papers each), based in Italy, who have developed statistical decomposition models for airport service performance [14, 15]. They are followed by Bellizzi, Park, and Zhang (4 papers each), whose work spans Italian airport LOS modelling, Korean/Australian passenger perceptions, and online review data mining, respectively. Figure 4 illustrates the top journals by publication count, Figure 5 the distribution of research methodologies, Figure 6 the most prolific authors, and Figure 7 the top contributing countries.

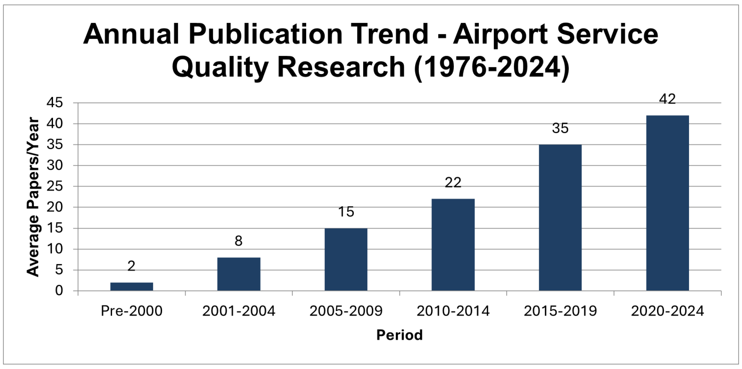


**Figure 2.** Annual publication trend in ASQ research by period (1976–2024)

Note: Values are approximate period averages. Annual growth rate = 8.25% (1976–2024); ASQ—airport service quality.

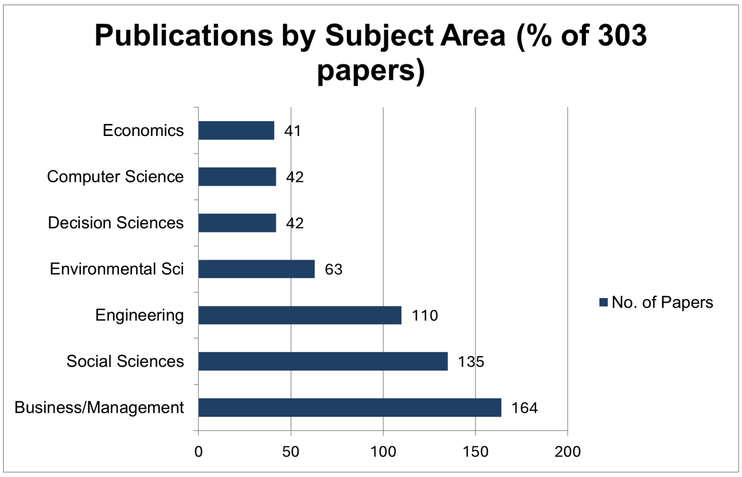


**Figure 3.** Distribution of ASQ publications by subject area (Scopus classification)

Note: Papers may belong to multiple subject areas, so percentages sum to >100%; ASQ—airport service quality.

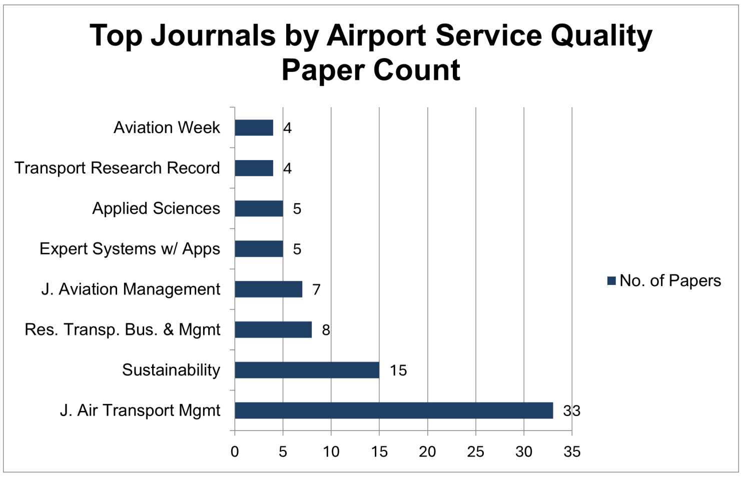


**Figure 4.** Top journals publishing ASQ research (1976–2024)

Note: ASQ—airport service quality.

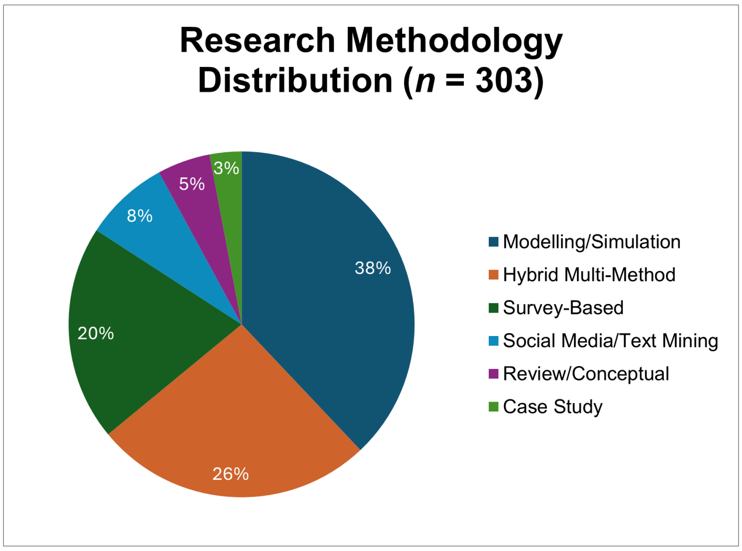


**Figure 5.** Distribution of research methodologies in the ASQ corpus

Note: ASQ—airport service quality.


**Figure 6.** Most prolific authors in ASQ research (1976–2024)

Note: ASQ—airport service quality.

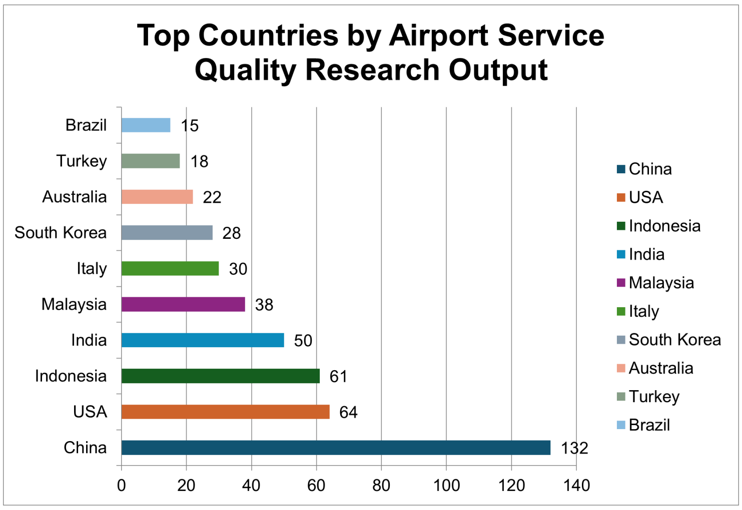


**Figure 7.** Top 10 countries by volume of ASQ research output (1976–2024)

Note: ASQ—airport service quality.

## 4 Airport Service Domains: Taxonomic Framework

The classification of airport service encounters into coherent domains underpins virtually every measurement instrument in the ASQ literature. This section traces the evolution of domain taxonomies from early operational distinctions to the multi-layered frameworks currently in use, and examines how each domain relates to the physical terminal layout and the passenger journey stages.

### 4.1 The Processing vs. Non-Processing Distinction

The most influential domain framework in ASQ literature is the processing/non-processing distinction introduced by Popovic et al. [3] and formalised by Kirk et al. [16] through the Taxonomy of Passenger Activities. Processing domains are mandatory for all passengers—check-in, security screening, immigration/customs, boarding—and create unavoidable service encounters with clear efficiency benchmarks. Non-processing domains are discretionary—retail, (F&B), lounges, rest areas, wayfinding—and primarily determine the experiential and emotional quality of the airport visit.

Correia et al. [2, 5] developed an early LOS framework distinguishing individual processing terminal components and providing aggregate performance measures. The significance of this distinction was reinforced by Bezerra and Gomes [17], who found that processing and non-processing activities have asymmetric effects on overall satisfaction —processing failures (long queues, poor check-in) cause strong dissatisfaction [18], while non-processing improvements (better retail, comfortable seating) add incremental satisfaction. This aligns with the Kano model's Must-Be versus attractive classification [12] and was subsequently confirmed by Fakfare and Wattanacharoensil [19] in the Thai airport context, consistent with asymmetric satisfaction patterns documented in broader hospitality and travel service research [20].

Bogicevic et al. [21] provide the most direct empirical test of this distinction, finding that processing and non-processing service quality affect passenger affective image and satisfaction through different mechanisms. Processing quality primarily reduces negative affect (dissatisfaction), while non-processing quality enhances positive affect (delight) [22], confirming the asymmetric satisfaction model in an airport-specific context, in line with broader evidence that nonlinear and asymmetric structures better capture real consumer satisfaction behaviour [23].

### 4.2 Stage-Based Domain Classification

Passenger travel through an airport is organised into three stages: departure, transit, and arrival. Each stage involves a distinct sequence of domain interactions. Most ASQ research focuses on the **departure stage** (approximately 78% of survey-based studies), with arriving-passenger studies accounting for fewer than 15% of papers, and transit-specific studies making up the remainder, as detailed in Table 3. This bias is partly methodological - departure lounges offer a captive, accessible population for intercept surveys, and partly a reflection of the industry's historical emphasis on departing passenger experience [24, 25].

**Table 3.** Stage-based domain mapping—departure, transit, and arrival stages

| Stage | Airport Domain | Domain Type | Key Service Factors | Primary KPIs |
|---|---|---|---|---|
| **Departure** | Kerbside/Curbside Access | Non-processing | Drop-off access, signage, taxi/rideshare, bus links | Vehicle dwell time (min); congestion index; walking distance to terminal (m) |
| | Check-in Hall | Processing | Queue management, self-check-in kiosks, baggage drop, staff efficiency | Average queue wait time (min); kiosk availability ratio; counter utilisation (%) |
| | Security Screening | Processing | Wait time, lane availability, staff behaviour, thoroughness | Average wait time (min); throughput rate (pax/hr); complaint rate per 1000 pax |
| | Immigration (departures) | Processing | Queue management, e-gate availability, officer attitude | Processing time (min); e-gate utilisation (%); queue length |
| | Departure Lounge/Gates | Non-processing | Seating, cleanliness, retail, F&B, Wi-Fi, ambience, wayfinding | Seat availability ratio; cleanliness inspection score; Wi-Fi speed (Mbps) |
| | Boarding Gate | Processing | Boarding efficiency, PA clarity, gate staff, gate change management | On-time boarding rate (%); gate change frequency; passenger confusion rate |

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| Stage | Airport Domain | Domain Type | Key Service Factors | Primary KPIs |
|---|---|---|---|---|
| **Transit/** Transfer | Transfer Corridor/ APM | Processing | Signage, APM frequency, walking distance, transit time | MCT compliance rate; APM headway (min) |
| | Transit Lounge | Non-processing | Seating, F&B access, rest rooms, immigration-free transit | Seat availability; rest room cleanliness score; complaint rate |
| | Customs/Security (re-entry) | Processing | Efficiency, clarity of procedures | Processing time; queue length |
| **Arrival** | Immigration (arrivals) | Processing | Queue management, e-gate availability, officer attitude | Average processing time (min); queue length; e-gate availability (%) |
| | Baggage Claim | Processing | Belt availability, delivery time, trolley access, lost baggage handling | First-bag delivery time (min); lost/damaged baggage rate per 1000; trolley availability ratio |
| | Customs (arrivals) | Processing | Channel clarity, officer attitude, declaration process | Processing time; queue management score |
| | Arrivals Hall/ Meet and Greet | Non-processing | Signage, information, F&B, money exchange, meet-and-greet | Information desk staffing hours; signage legibility score |
| | Landside Egress (Arrivals) | Non-processing | Taxi/rideshare access, pre-paid taxi, bus links, car park, hotel shuttle | Taxi wait time (min); transport variety index; car park proximity (m) |

Note: KPIs—key performance indicators; F&B—food and beverage; PA—public address; APM—airport people mover; MCT—minimum connection time.

### 4.3 Airside vs. Landside

A persistent gap in ASQ research is the neglect of landside service quality. Airside domains (post-security) dominate the literature, accounting for over 80% of attribute-level studies. Landside domains—kerbside access, car parking, ground transportation, pre-arrival online check-in, and public transport connections—constitute the first and last touchpoints of the passenger journey, yet are studied in fewer than 12% of papers [5]. The notable exception is wayfinding and signage research in landside access contexts.

The Indian context makes this gap particularly pressing. As Indian airports undergo rapid terminal expansion (Delhi T3, Mumbai T2, Bengaluru T2, Hyderabad), the landside infrastructure—city-side roads, metro connections, bus services, taxi/app-cab management—has emerged as a critical constraint on overall passenger experience. The airport authority of India (AAI) ACI ASQ data confirms that landside-related attributes consistently underperform in Indian airport benchmarking [26].

### 4.4 Functional Efficiency: Level of Service-Based Domain Assessment

Correia et al. [2] proposed a global index for LOS evaluation at airport passenger terminals, integrating individual domain scores into an aggregate terminal performance measure using both objective data (queue lengths, waiting times, floor space per passenger) and subjective passenger ratings. This approach was extended by Allen et al. [25] and Correia et al. [5], who decomposed LOS at a mid-sized Italian terminal into individual component scores, accounting for sociodemographic heterogeneity using a structural equation modeling-multiple indicators and multiple causes (SEM-MIMIC) ordinal probit model. The ACRP Report 25 [27, 28] provides the most comprehensive objective LOS framework for airport terminal planning and design, specifying Level A (excellent) through Level F (unacceptable) standards for each processing and non-processing domain based on space, queue, and time benchmarks.

## 5 Service Dimensions, Parameters and Key Performance Indicators

While domains describe where service encounters take place, dimensions describe the qualitative lenses through which passengers evaluate those encounters. This section examines how generic service quality dimensions have been adapted for the airport context, how the ACI ASQ programme operationalises these dimensions, and how the literature maps specific measurable attributes and key performance indicators (KPIs) to each domain-dimension intersection.

### 5.1 SERVQUAL-Derived Dimensions in the Airport Context

Parasuraman et al.'s [4, 8] SERVQUAL model—measuring five dimensions (Tangibles, Reliability, Responsiveness, Assurance, Empathy) through gap scores between expectations and perceptions—was the dominant framework for early ASQ measurement. A sizeable number of studies have adapted SERVQUAL for airports [4, 9, 29]. However, the expectations-minus-perceptions gap approach has been critiqued for the airport context because passengers, especially first-time travellers to unfamiliar airports, may have poorly formed a priori expectations. This led to the adoption of SERVPERF [9], which measures only performance perceptions, in many airport studies [30].

Fodness and Murray [1] identified airport-specific service quality attributes through passenger interviews and observations, finding that the standard SERVQUAL dimensions required substantial adaptation for the airport context. Their work, alongside that of Bezerra and Gomes [17] and the ASQ Systematic Review, established that the five SERVQUAL dimensions, while directionally relevant, underrepresent the specificity of airport service encounters - particularly the sequential, stage-based nature of the airport journey and the coexistence of processing and non-processing activities within the same physical space.

### 5.2 Airports Council International Airport Service Quality Dimensions as the Industry Standard

The ACI ASQ Programme, documented extensively in the 2012 ACI (ACI Airport Performance Measures Guidebook) [31] and the 2023 ACI [32], provides the most widely used standardised framework for measuring passenger satisfaction globally [33]. Operating across 340+ airports and covering approximately 85% of global air traffic, the ACI ASQ programme administers approximately 650 passenger surveys per quarter per participating airport in departure lounges, covering 34 attributes grouped under six broad categories. The six ACI ASQ categories are: (1) Access, (2) Check-in, (3) Security/Passport Control, (4) Airport Environment, (5) Food, Beverages, and Shopping, and (6) Airport Services. ACI releases quarterly rankings, annual World Airport Traffic Reports, and regional performance comparisons—making it the de facto global benchmark for airport passenger satisfaction. Table 4 presents the ACI ASQ survey categories and key attributes.

**Table 4.** ACI airport service quality survey categories and key attributes

| ACI Category | Key Attributes Measured | Average Importance Weight (ACI) | Corresponding Research Domain |
|---|---|---|---|
| Access | Ground transportation to/from airport; car parking; walking distance; signage approaching airport | High | Landside/kerbside |
| Check-in | Queue length at check-in; courtesy and efficiency of check-in staff, self-service kiosk availability | Very high | Check-in Hall |
| Security/passport control | Waiting time at security; courtesy of security staff; thoroughness; immigration wait time | Very high | Security/immigration |
| Airport environment | Cleanliness of terminal; comfort of gate seating areas; ambience; temperature and ventilation; washroom cleanliness | High | Terminal/departure lounge |
| Food, beverages & shopping | Value for money of F&B; variety of F&B outlets; variety of shops; tax-free shopping | Moderate | Retail/F&B |
| Airport services | Courtesy and helpfulness of airport staff; availability of baggage carts/trolleys; Wi-Fi; flight information screens; wayfinding signage | High | All domains |

Note: ACI—airports council international; F&B—food and beverage.

### 5.3 Mapping the 34-Attribute Framework to Domains and Key Performance Indicators

Across the 303-paper corpus, 34 distinct service attributes were consistently identified and studied. The comprehensive mapping below integrates domain classification, performance indicators, passenger importance weights, and the symmetry of their effect on satisfaction (Kano classification), providing the most complete cross-referenced attribute framework available in the literature (Table 5).

**Table 5.** Comprehensive service attribute–domain–Key Performance Indicator (KPI) mapping

| Attribute | Domain | Stage | Key KPIs | Mean Wt. | Research Coverage | Kano Type |
|---|---|---|---|---|---|---|
| Kerbside/curbside facilities | Landside | Dep/Arr | Vehicle dwell time; congestion score; drop-off distance | 1.00 | 6.9% | Must-Be |
| Cleanliness of washroom facilities | Washrooms | All | Cleaning cycle frequency (min); complaint rate; hygiene score | 0.87 | 6.9% | Must-Be |
| Information/wayfinding (general) | Wayfinding | All | Lost-passenger rate; information desk response time; signage legibility score | 0.83 | 20.7% | 1D |

*Continued on next page*

| Attribute | Domain | Stage | Key KPIs | Mean Wt. | Research Coverage | Kano Type |
|---|---|---|---|---|---|---|
| Processing times | Check-in, security, or immigration | Dep/Arr | Average queue wait time (min); peak queue length; throughput rate | 0.83 | 17.2% | Must-Be |
| FIDS-flight information screens | Departure lounge/gates | All | Screen update frequency; error rate; viewing angle coverage (%) | 0.83 | 37.9% | 1D |
| Washroom and shower facilities | Washrooms | All | Availability ratio (no. per 1000 pax); shower booking time; cleanliness rating | 0.82 | 24.1% | Must-Be |
| Check-in facilities | Check-in hall | Departure | Wait time; kiosk availability; staff courtesy score; bag drop efficiency | 0.81 | 31.0% | 1D |
| Security screening | Security lane | Departure | Wait time; lane availability; staff behaviour score; throughput | 0.79 | 24.1% | Must-Be |
| Airline information counter | Check-in hall | Departure | Counter availability; response accuracy; staff knowledge score | 0.78 | 6.9% | 1D |
| Baggage delivery times | Baggage claim | Arrival | First-bag delivery time (min); last-bag delivery time; carousel wait | 0.74 | 17.2% | Must-Be |
| Terminal signage (boarding/transfer/arrivals) | Wayfinding | All | Signage consistency score; multilingual coverage; update frequency | 0.73 | 37.9% | 1D |
| People mover/APM system | Transfer/transit | Transit | Headway (min); capacity utilisation; reliability rate (%) | 0.71 | 10.3% | Must-Be |
| Lounges | Departure lounge | Departure | Access ratio (% eligible pax); capacity utilisation; satisfaction score | 0.69 | 13.8% | Attractive |
| Clarity of boarding calls/PA system | Gate/departure lounge | Departure | Intelligibility score; complaint rate; language coverage | 0.68 | 24.1% | Must-Be |
| ATM/banking facilities | Departure lounge/arrivals | All | ATM availability ratio; queue length; functional uptime (%) | 0.65 | 13.8% | Attractive → 1D |
| Staff attitude and courtesy | All domains | All | Mystery shopper score; complaint rate per 1000 pax; satisfaction rating | 0.65 | 20.7% | 1D |
| Lifts/escalators/moving walkways | Terminal | All | Operational uptime (%); complaint rate; coverage (% of vertical transitions) | 0.64 | 17.2% | Must-Be |
| Parking facilities | Landside | Dep/Arr | Availability ratio; walking distance; cleanliness; pricing satisfaction | 0.62 | 10.3% | 1D |
| Choice of shopping (tax-free) | Retail | Departure | Spend per passenger (INR/EUR); outlet variety index; brand tier coverage | 0.61 | 27.6% | Attractive |
| Choice of F&B (Bars, Cafes, Restaurants) | F&B | All | F&B spend per passenger; value-for-money score; outlet variety index | 0.60 | 27.6% | Attractive |
| Seating facilities throughout terminal | Departure lounge | All | Seat availability ratio (seats per 1000 pax-hour); charging point ratio | 0.59 | 20.7% | Must-Be |
| Ease of transit through airport | Transit/transfer | Transit | Minimum connection time compliance; signage legibility; transfer time | 0.59 | 31.0% | 1D |
| Luggage trolleys (airside and landside) | Baggage/landside | Arr/Dep | Trolley availability index; return compliance rate; coverage area | 0.59 | 17.2% | Must-Be |
| Terminal cleanliness (general) | Terminal | All | Third-party cleanliness audit score; complaint rate; cleaning cycle time | 0.58 | 13.8% | Must-Be |
| Internet/Wi-Fi availability | All domains | All | Coverage (% of terminal); speed (Mbps); connection success rate | 0.57 | 10.3% | Attractive → 1D |
| Terminal comfort, ambience, and design | Terminal | All | Ambient temperature (°C range); noise level (dB); aesthetic satisfaction score | 0.48 | 20.7% | Attractive |
| Children's play area and facilities | Departure lounge | All | Availability (binary); area per child passenger; safety inspection frequency | 0.42 | 10.3% | Attractive |

Note: KPIs—Key Performance Indicators; FIDS—Flight Information Display Systems; APM—Airport People Mover; ATM—Automated Teller Machine; PA—Public Address; F&B—Food and Beverage.

Source: Synthesised from SLR corpus (303 papers, 1976–2024); ACI [31]; ACRP [27]; ACRP (2012) [28]. Mean weights derived from cross-study synthesis (see Paper 2 for methodology details).

## 6 Computing the Airport Service Quality Score: From Attributes to Terminal to Airport Index

The aggregation of individual service attribute ratings into a domain score, a terminal score, and ultimately a composite ASQ index is a critical methodological step that is often underspecified in the academic literature. This section synthesises evidence on how different studies have approached the computation of ASQ scores across domains, journey stages, and multi-terminal airports.

### 6.1 Domain-Level Scoring

At the most granular level, ASQ studies measure individual service attributes on a Likert or semantic differential scale (typically 1–5 or 1–7). Domain-level scores are then computed by one of three approaches:

- **Simple arithmetic mean**: The domain score is the unweighted average of all attribute ratings within the domain. This is the most commonly used approach for its transparency and ease of communication [2, 32].
- **Weighted mean**: Attributes are weighted by importance scores derived from regression coefficients, Importance-Performance Analysis (IPA) quadrant analysis, or expert elicitation. The domain score is the sum of (weight × rating) across attributes. This approach is used by Eboli and Mazzulla [14, 15], Allen et al. [25], and Fakfare and Wattanacharoensil [19].
- **Latent factor score**: In SEM-based studies, domain scores are latent constructs estimated by the model parameters rather than simple averages. The latent score accounts for measurement error in individual items. This is the most technically rigorous approach but requires confirmatory factor analysis [17].

### 6.2 Stage-Based Aggregation

ASQ encompasses multiple journey stages—departure, transit, and arrival—each with distinct service processes and attributes. Evidence from the corpus shows strong concentration on the departure stage:

- **Departure-only studies** constitute approximately 70–75% of the corpus, consistent with the ACI ASQ Programme's traditional focus on departing passengers surveyed in the departure lounge [31, 32].
- **Arrival-only studies** are rare (about 5% of the corpus), despite the different service experience faced by arriving passengers (customs, baggage reclaim, ground transportation).
- **Multi-stage studies** covering both departure and arrival—and occasionally transit—represent approximately 20–25% of the corpus but rarely compute a unified cross-stage ASQ score. Instead, they compare stage-specific satisfaction ratings side by side [5, 29].

A unified multi-stage ASQ index requires a decision on stage weighting: Is an arriving passenger's experience equal in weight to a departing passenger's? In high-traffic hub airports where many passengers are in transit, a transit-weighted composite index may be more representative. The ACI ASQ Programme does not publicly report a cross-stage composite; stage-specific scores are reported separately.

### 6.3 Terminal-Level vs. Airport-Level Scoring

Multi-terminal airports present a particular aggregation challenge. Terminals at major airports often differ substantially in design vintage, capacity, operating airlines, and passenger mix—making a single airport-level score potentially misleading. The literature and industry approaches handle this in three ways:

- **Single-terminal studies**: Most academic studies focus on one terminal, typically the primary international departure terminal, and report a single terminal-level ASQ score. This is operationally tractable but limits generalisability to multi-terminal airports.
- **Airport-level blended scores**: The ACI ASQ Programme reports a single airport-level score, averaging survey responses across all terminals where surveys are deployed. This is practical for comparative benchmarking but masks within-airport variation.
- **Terminal-specific comparative analysis**: A small number of studies explicitly compare terminal scores. Examples include studies at London Heathrow (T2 vs. T5), Dubai (T1 vs. T3), and Delhi (T1 vs. T3) that demonstrate significant inter-terminal variation in passenger satisfaction, particularly for check-in efficiency and retail quality (Table 6).

**Table 6.** Multi-domain/multi-terminal aggregation approaches in selected ASQ studies

| Study | Airport(s) | No. Terminals | Stages Covered | Aggregation Method | Terminal Scores Separated? |
|---|---|---|---|---|---|
| [32] | 300+ airports globally | All (blended) | Departure only | Simple average of attribute ratings; airport-level composite | No (airport-level only) |
| [2, 5] | Multiple international | 1–2 per airport | Departure + arrival | Weighted mean (regression-derived weights) | No |
| [14, 15] | Italian airports | 1 | Departure | SEM latent scores + statistical decomposition | N/A (single terminal) |
| [25] | Rome Fiumicino | 2 | Departure | Weighted mean (IPA-derived importance weights) | Yes (terminal comparison) |
| [17] | Brazilian airports | 1 | Departure | CFA/SEM latent factor model | N/A (single terminal) |
| [21] | USA airports | Multiple | Departure | Text mining/sentiment aggregation | No |
| [23] | Asian hub airport | 1 | Departure | SEM + BN + ANN hybrid [34] | N/A (single terminal) |
| [27, 28] | US airports | Multiple | Departure + arrival | Simple average; domain-level reporting | No (airport-level) |
| [19] | Thai airport | 1 | Departure | IPA + Kano integrated weighting | N/A (single terminal) |
| [35] | Korean airports | 1 | Departure | service performance weighted mean | N/A (single terminal) |

Note: ASQ—airportservice quality; SEM—structural equation modeling; IPA—importance-performance analysis; CFA—confirmatory factor analysis; BN—bayesian network; ANN—artificial neural network; N/A—not applicable.

### 6.4 Composite Airport Service Quality Index Construction

A composite airport service quality index (ASQI) aggregates domain-level scores into a single numeric indicator of overall ASQ. Three main approaches are used:

**Weighted Sum Formula:**

$$\mathrm{ASQI} = \Sigma\,(\mathrm{w}_d \times \mathrm{S}_d) \tag{1}$$

where, $\mathrm{w}_d$ is the relative importance weight of domain d and $\mathrm{S}_d$ is the mean satisfaction score for domain $d$.

Weights can be derived from regression analysis (standardised beta coefficients), importance ratings (IPA) [36], Eboli and Mazzulla [14] proposed a methodology for evaluating hybrid service quality by integrating both subjective (passenger-perceived) and objective (operational) measures, offering a multi-dimensional assessment framework [14, 15]; Liou et al. [30] use Analytic Hierarchy Process (AHP); Allen et al. [25] use IPA importance ratings.

**Principal Component Analysis-Based Index:**

$$\mathrm{ASQI} = \Sigma\,(\lambda_k/\Sigma\lambda \times \mathrm{PC}_k) \tag{2}$$

where, $\lambda_k$ is the eigenvalue of the $k$-th principal component.

This approach, used in some composite index studies, lets the data determine the weighting structure through principal component analysis. The first principal component typically explains 40–60% of total variance in airport attribute ratings and serves as the index [17].

**Confirmatory Factor Model-Based Index:**

$$\mathrm{ASQI} = \frac{\sum_{i,j=1}^{n}(\text{factor loading}_{ij} \times \text{attribute rating}_{ij})}{\text{no. of attributes in domain}} \tag{3}$$

In Confirmatory Factor Analysis (CFA)/SEM-based studies, the composite index is derived from the measurement model, using standardised factor loadings as de facto attribute weights within each domain. This is the most statistically rigorous approach [17].

### 6.5 Implications for the Indian Airport Context

Major Indian international airports (Delhi IGI, Mumbai CSIA, Bangalore, Hyderabad, Chennai) are all multi-terminal airports with significant variation in terminal age, design, and passenger profile. Delhi IGI operates three active terminals (T1, T2/T3 combined), each serving different airline categories. A robust ASQI for Indian airports must therefore address:

- Terminal-level data collection (not airport-aggregated)
- Stage disaggregation (at minimum, departure vs. arrival separation)
- Appropriate weighting of domains for the landside-heavy Indian airport experience (where pre-security areas, parking, and access are significant pain points)
- Objective data integration alongside passenger perception scores

## 7 Passenger Profiling: Heterogeneity in Service Quality Perceptions

One of the most consistent findings in ASQ literature is that passengers do not constitute a homogeneous group. Perceptions of service quality, importance weights assigned to attributes, and the relationship between attribute performance and overall satisfaction vary substantially across demographic, travel, and cultural dimensions. This section reviews the evidence by profile category.

### 7.1 Demographic Variables

**Gender:** The moderating role of gender on ASQ perceptions has been studied extensively [17]. Women consistently report higher sensitivity to washroom cleanliness, safety, and security, and staff courtesy, while men demonstrate higher importance weights for efficiency-related attributes (check-in speed, processing time) and retail variety. Bezerra and Gomes [17] specifically examined the moderating effect of gender on retail patronage intentions at airports, finding that women assign significantly higher importance to the shopping environment, product range, and staff service, whereas men are more price-sensitive. These findings have practical implications for airport retail strategy and terminal design—specifically, the allocation of washroom space and the zoning of retail areas [37].

**Age:** Older passengers (60+) consistently rate accessibility features, staff assistance, seating availability, and clear public address systems as most critical [5, 25]. They demonstrate lower tolerance for processing delays and higher sensitivity to wayfinding confusion. Younger passengers (18–35) prioritise digital services—Wi-Fi quality, self-service kiosks, FIDS accuracy, and mobile boarding pass scanning efficiency [38]—and are the dominant users of social media feedback channels [7]. Middle-aged passengers (36–55), particularly business travellers, show the highest sensitivity to lounge access, fast-track security lanes, and on-time departure reliability. Allen et al. [25] explicitly modelled age-group heterogeneity using an MIMIC ordinal probit approach, confirming statistically significant differences across age cohorts in a mid-sized Italian terminal.

**Education and Income:** Higher-educated and higher-income passengers tend to have higher absolute expectations, resulting in lower satisfaction scores for equivalent service levels [17]. However, they also demonstrate greater capacity to contextualise and differentiate between attributes, resulting in more nuanced satisfaction profiles. Premium cabin passengers exhibit significantly higher expectations for lounge quality, fast-track processing, and personalised service [1, 35]. Table 7 summarises the demographic profile effects on key ASQ attributes.

**Table 7.** Passenger demographic profile effects on key ASQ attributes

| Profile Variable | Attributes with Higher Importance | Attributes with Lower Importance | Key References |
|---|---|---|---|
| Female passengers | Washroom cleanliness, safety/security, staff courtesy, children's facilities, seating comfort | Retail variety, F&B choice, processing speed | [17, 24, 25, 37] |
| Male passengers | Processing efficiency, checkin speed, parking, retail, F&B | Washroom facilities, ambient comfort | [17, 25] |
| Younger (18–35) | Wi-Fi, self-service kiosks, FIDS accuracy, charging points | Staff assistance, PA clarity, accessibility | [7, 21, 38, 39] |
| Older (60+) | Staff assistance, seating, PA clarity, accessibility (lifts/escalators), signage | Wi-Fi, self-service, retail | [5, 17, 18, 25] |
| Business travellers | Lounge access, fast-track security, processing time, FIDS, on-time departure | Retail variety, children's facilities, ambience | [1, 21, 35] |
| Leisure travellers | F&B variety, retail, terminal ambience, children's facilities, comfort seating | Lounge, fast-track, parking pricing | [17, 19, 35] |
| Frequent flyers ($\geq$10 flights/year) | Processing efficiency, lounge, FIDS, Wi-Fi, staff professionalism | Ambience, children's facilities, first impressions | [1, 21, 25] |
| First-time visitors | Wayfinding, signage clarity, information desks, staff helpfulness | Lounge, fast-track, parking | [1, 17, 21] |

Note: ASQ—airport service quality; F&B—food and beverage; FIDS—flight information display systems; PA—public address

### 7.2 Passenger Type: Departing, Arriving, and Transit Passengers

The overwhelming majority of ASQ research focuses on departing passengers. Studies of arriving passengers are rare, despite the arrival experience constituting the first on-ground impression for incoming tourists and the final impression for outbound travellers. The few studies that have compared departing and arriving passengers consistently find different priority structures: arriving passengers weight baggage delivery speed, immigration processing time, and landside exit accessibility most highly, while departing passengers weight check-in efficiency, security processing, and departure lounge comfort [21].

**Transit passengers** constitute a distinct category with unique needs: minimum connection time reliability, clear transfer signage, transit lounge facilities, and immigration-free transfer procedures. The Jakarta Airport People Mover (APM) study specifically addressed transit passenger ASQ, finding that APM reliability and frequency had the strongest effect on transfer satisfaction. evaluated LOS specifically for transfer passengers, establishing separate LOS benchmarks for transfer corridors, re-security screening, and gate-to-gate transit time.

### 7.3 Domestic vs. International Passengers

International passengers demonstrate significantly different ASQ priorities compared to domestic passengers. Key differences documented in the literature include:

- **Immigration and customs processing:** Critical only for international passengers; accounts for approximately 15–20% of total dissatisfaction variance in international airport studies [17].
- **Multilingual signage and information:** International passengers—particularly non-English speakers—show higher sensitivity to wayfinding clarity and staff language competence.
- **Currency exchange and banking:** More important for international passengers, especially those arriving from countries with currency controls.
- **Shopping and duty-free:** International departing passengers show substantially higher retail spending propensity and importance weights for duty-free variety.
- **Domestic passengers:** More sensitive to check-in speed, gate proximity, and parking—reflecting shorter pre-flight time budgets and familiarity with the airport environment [26].

In the Indian context specifically, ASQ Literature Study [26] identifies that Indian domestic passengers prioritise check-in speed, cleanliness, and ground transportation access, while international passengers at Indian airports place higher priority on immigration processing, duty-free shopping, and terminal ambience.

### 7.4 Travel Class Effects

Travel class—economy, business, and first/premium—is a significant moderator of ASQ perceptions. Business and first-class passengers consistently report higher pre-flight lounge usage, higher expectations for fast-track security lanes, and greater sensitivity to personalised service [35]. They demonstrate lower tolerance for general service failures (e.g., security queue length) but are more forgiving of non-processing shortcomings when compensated by lounge quality. Economy passengers, particularly LCC travellers, exhibit a distinct pattern: lower baseline expectations, higher sensitivity to value-for-money in F&B, and greater importance placed on FIDS accuracy and seating availability [19, 40].

## 8 Geographic and Cultural Variations in Airport Service Quality Perceptions

Geography and culture shape both the supply side (how airports are designed and operated) and the demand side (what passengers expect and how they evaluate service) of ASQ. Table 8 summarises geographic patterns in ASQ research by region. ASQ literature study (exploring different nationality perceptions of ASQ) provides direct comparative evidence, finding statistically significant differences in service quality perceptions across passenger nationalities at the same airport [41]. Cultural dimensions (power distance, uncertainty avoidance, individualism/collectivism from Hofstede [42]) partially explain these differences.

**Table 8.** Geographic patterns in ASQ—regional focus, findings, and key studies

| Region | Research Volume | Dominant Methods | Key Findings | Key Papers |
|---|---|---|---|---|
| **East Asia** (China, Korea, Taiwan, Japan) | High (China: 132 papers) | Big data, sentiment analysis, ANN, fuzzy MCDM | High focus on digital services, self-service technology, social media ASQ; Incheon Airport consistently top-ranked; Taiwan airports widely studied with fuzzy methods | [7, 30, 34, 43] |
| **Southeast Asia** (Indonesia, Malaysia, Thailand, Singapore) | High (Indonesia: 61, Malaysia: 38) | SEM, IPA, SERVQUAL, survey-based | LCC airport-specific studies (KLIA2); technology adoption at kiosks; strong focus on loyalty and eWORM: Thai asymmetric attribute classification | [19, 30, 40] |
| **Middle East** (UAE, Saudi Arabia) | Moderate | Survey-based, case study, benchmarking | Dubai International consistently ranked high [44]; innovation and luxury attributes studied; cultural service expectations explored | [21, 44, 45] |
| **Europe** (Italy, UK, Germany, Spain) | Moderate | Statistical models, SEMMIMIC, IPAGap, DEA | Italian researchers dominate (Eboli/Mazzulla); level-of-service frameworks; sustainability benchmarking; heterogeneity modelling | [14, 15, 24] |
| **Americas** (USA, Brazil) | Moderate (USA: 64, Brazil: 15) | ACRP frameworks, regression, foundational frameworks | US airports studied through ACRP/ J.D. Power lens; Brazilian airports via regressionbased IPA; foundational SERVQUAL work originated here | [6, 27, 28] |
| **Africa** (Nigeria) | Low | IPA, surveybased | Lagos Airport studied through IPA [46]; service quality found significantly below expectations; infrastructure deficit documented | [17, 46, 47] |
| **Oceania** (Australia) | Low–Moderate | Survey-based, case study | Melbourne Airport passenger experience [48]; complaint intention modelling; Korean vs Australian comparison | [35, 48, 49] |
| **South Asia** (India) | Low (contextspecific) | Survey, DEA, integrated performance evaluation | Most Indian-authored papers cover foreign airports; Indian-specific studies limited to Delhi, Mumbai; AAI/ACI data underanalysed; PPP framework studied [50] | [26, 31, 50] |

Note: ANN—artificial neural network; MCDM—multiple-criteria decision making; ASQ—airport service quality; SEM—structural equation modeling; IPA—importance-performance analysis; SERVPERF—service performance; LCC—low-cost carrier; eWOM—electronic word-of-mouth; MIMIC—multiple indicators and multiple causes; DEA—data envelopment analysis; ACRP—airport cooperative research program; AAI—airport authority of India; ACI—airports council international; PPP—public-private partnership

Cultural response style is a well-documented confound in international ASQ comparisons. Asian passengers tend to use the central region of rating scales more frequently (acquiescence bias), leading to compressed score distributions that may mask genuine performance differences [51]. European and North American passengers tend toward more extreme ratings. This makes direct cross-cultural comparisons of mean satisfaction scores unreliable without statistical correction, a methodological limitation acknowledged in the literature [17, 51].

## 9 Survey Methodologies in Airport Service Quality Research

The validity and generalisability of ASQ findings depend critically on how passenger data are collected. This section reviews the dominant survey approaches in the literature, examines known biases that affect airport-based data collection, and benchmarks typical sample sizes against the requirements of common analytical methods.

### 9.1 Survey Design and Instrument Characteristics

The questionnaire remains the dominant data collection instrument in ASQ research, used in approximately 74% of empirical studies (survey-based: about 20%; survey + other methods: about 54%). Most questionnaires employ a **5-point Likert scale** for both importance and performance ratings, though some studies use 7-point scales for greater discrimination [27, 28].

ASQ literature study provides one of the most comprehensive guides to questionnaire design for airport passenger surveys, detailing the cognitive burden of different attribute list lengths, the optimal sequencing of importance and performance questions, the handling of multi-leg journeys, and the design of airport-specific contextual anchors. Key design considerations, summarised in Table 9, include: (1) keeping attribute lists to 20–35 items to avoid respondent fatigue; (2) using specific, observable attribute descriptions rather than abstract dimension labels; (3) placing overall satisfaction questions after attribute ratings to avoid anchoring bias; and (4) collecting passenger profile data at the end of the questionnaire.

**Table 9.** Survey methodology characteristics in ASQ research

| Design Parameter | Common Practice in Literature | Range | ACI ASQ Standard |
|---|---|---|---|
| Rating scale | 5-point Likert (most common); 7-point (some studies) | 3–10 point scales | 5-point (1 = Excellent, 5 = Poor) |
| Number of attributes | 20–35 | 8–74 | 34 attributes |
| Sample size (inperson) | 200–500 per study | 50–1500 | ∼650 per quarter per airport |
| Sample size (online reviews) | 1,000–50,000 reviews | 500–200,000+ | N/A (proprietary) |
| Survey location | Departure lounge (postsecurity), gate areas | Check-in, lounge, gate, arrivals | Departure lounge (international/domestic) |
| Survey timing | Post-check-in, pre-departure (most common) | Pre-/postsecurity; arrivals; online | Post-security departure lounge |
| Administration method | Self-administered; intercept interviews | CAPI, paper, tablet, online | Face-to-face intercept (trained enumerators) |
| Survey language | Primarily English; some local language versions | 1–8 languages | Multiple languages (airportspecific) |
| Duration | 10–15 mins | 5–30 mins | Approximately 10 mins |
| Sampling method | Convenience/systematic (every nth passenger) | Convenience, stratified, quota | Quota sampling by flight (origin/destination/airline) |

Note: ACI—airports council international; ASQ—airport service quality; CAPI—computer-assisted personal interviewing; N/A—not applicable

### 9.2 Biases and Limitations in Survey-Based Airport Service Quality Research

Survey-based methods face several well-documented biases in the airport context. The **captive audience effect**—surveying passengers in departure lounges who have little else to do—may inflate response rates but also introduces selection bias, as passengers with very long waits may have more negative general affect influencing all attribute ratings [10]. The **halo effect** is particularly pronounced in airport surveys: passengers who experience a flight delay (airline responsibility) tend to rate airport service attributes lower, even unrelated ones such as washroom cleanliness. The **primacy/recency effect** means that the first (kerbside) and last (boarding) touchpoints disproportionately influence overall satisfaction [48, 52], which has implications for resource allocation [3].

ASQ literature study (A measurement tool to determine the quality of passenger experience: subjective and objective functions) is one of the few studies that directly compares subjective passenger ratings with objective service performance data (measured queuing times, cleanliness inspection scores), finding moderate-to-high correlation ($r$

= 0.62–0.78) but with systematic underestimation of objective performance by subjective ratings in high-stress processing situations [53].

### 9.3 Average Sample Sizes by Study Type

In-person airport survey studies typically report sample sizes of 200–500 complete responses, with a median of approximately 350. Studies in Asian airports tend to report larger samples (Boonchunone et al. [54]: $n$ = 400; Usman et al. [55]: $n$ = 356; Fakfare et al. [19]: $n$ = 381) compared to European or Middle Eastern studies [25]. Online review studies report sample sizes in the thousands to hundreds of thousands: Alanazi et al. [7] (Skytrax analysis) used over 14,000 reviews; ASQ literature study (online reviews) used over 22,000 reviews; and ASQ literature study (sentiment analysis of airport websites) analysed over 8,500 comments. The ACI ASQ programme collects approximately 650 surveys per airport per quarter—approximately 2,600 per year—providing the largest standardised comparative dataset in the field.

## 10 Social Media and Digital Platforms in Airport Service Quality Research

The use of social media and online review platforms as data sources for ASQ research has grown dramatically from approximately 2015 onwards, paralleling the growth of digital travel platforms and the availability of big data analytical tools. This section reviews the major platforms and methodological approaches documented in the literature.

### 10.1 Skytrax—Structured Online Review Mining

Skytrax (worldairportreviews.com) is the largest dedicated airport review database, with a structured review format covering approximately 46 service criteria on a 5-star scale, with an overall star rating from 1 to 10. Alanazi et al. [7] provide the most thorough analysis of Skytrax data in the academic literature, analysing over 14,000 reviews across 30 airports. Key findings include: (1) terminal cleanliness and staff service are the strongest predictors of overall Skytrax star ratings; (2) Skytrax ratings show significant correlation ($r \approx 0.71$) with contemporaneous ACI ASQ scores for the same airports, suggesting reasonable convergent validity; and (3) Skytrax data exhibits selection bias toward very satisfied (5-star) and very dissatisfied (1–2 star) reviewers, with a bimodal distribution that differs from the roughly normal distribution of in-person survey data.

A critical advantage of Skytrax data is its availability across multiple years, enabling longitudinal analysis that is extremely rare in traditional survey-based ASQ research. (Has passenger satisfaction at airports changed?) exploited Skytrax time-series data to examine pre- and post-Coronavirus Disease (COVID) changes in ASQ [56], finding that cleanliness and social distancing management became the dominant satisfaction drivers during the 2020–2024 period, displacing retail and F&B, which had previously been more prominent.

### 10.2 Twitter/X—Real-Time Sentiment Analysis

Twitter ASQ Sentiment Study [57] and Online Review Mining Studies, 2020–2024 (Applying deep learning models to Twitter data to detect ASQ) [7] represent the emerging strand of Twitter/X-based ASQ research. Twitter data offers real-time, unsolicited feedback—passengers tweet during the service encounter rather than retrospectively—which may provide a more emotionally authentic and less cognitively filtered picture of service experience. However, Twitter data suffers from severe demographic selection bias (younger, more tech-savvy travellers; predominantly English-language), sentiment attribution challenges (negative tweets tend to go viral, creating negativity bias in raw sentiment counts), and the difficulty of airport-specific attribution versus airline attribution in combined journey tweets.

Online Review Mining Studies, 2020–2024 [7] (Social media as a resource for sentiment analysis of ASQ) provides an in-depth review of social media methodologies in ASQ research, including Twitter, Facebook, Instagram, and review platforms. The paper identifies three broad approaches: (1) lexicon-based sentiment scoring using pre-built dictionaries; (2) supervised machine learning classifiers; and (3) deep learning models (Bidirectional Encoder Representations from Transformers and Long Short-Term Memory). Deep learning approaches outperform lexicon methods on domain-specific airport language but require large labelled training datasets that are expensive to create.

### 10.3 Comparison of Data Sources: Traditional Survey vs. Social Media

ASQ studies primarily rely on two sources of passenger feedback: traditional surveys and social media data. While surveys provide structured and statistically reliable insights into passenger satisfaction, social media platforms offer large-scale, real-time, and unsolicited passenger opinions. Together, these approaches provide complementary perspectives for evaluating ASQ. Table 10 compares the key characteristics, strengths, and limitations of both data sources. Table 10 provides a systematic comparison of data collection methods used in ASQ research, highlighting the primary method, data sources, sample sizes, key findings, and limitations associated with each approach.

**Table 10.** Comparison of data collection methods for ASQ research

| Parameter | In-Person Intercept Survey | Online Questionnaire | Skytrax/Trip Advisor Reviews | Twitter/Social Media |
|---|---|---|---|---|
| Sample size | 200–600 (small) | 300–2000 (moderate) | 1,000–200,000+ (large) | 10,000–1,000,000+ (very large) |
| Demographic representativeness | Moderate (departure lounge bias) | Low (online/younger bias) | Low (dissatisfied/highly satisfied bias) | Very low (young, tech-savvy, English speaking) |
| Temporal coverage | Cross-sectional (typically 1 week–3 months) | Cross-sectional | Longitudinal (years of data available) | Real-time; near continuous |
| Attribute-level precision | High (structured questionnaire) | High (structured) | Moderate (semistructured review format) | Low (unstructured; requires NLP extraction) |
| Convergent validity with ACI ASQ | High ($r \approx 0.75$–$0.85$) | Moderate-high | Moderate ($r \approx 0.71$) | Low-moderate |
| Cost | High (trained surveyors, airport access) | Moderate (panel costs) | Low (scraping/API) | Low-moderate (API, NLP tools) |
| Key strengths | Rigorous, validated instruments; sampling control | Large geographic reach; lower cost | Longitudinal trends; large $n$; unprompted feedback | Real-time; spontaneous; emotional content |
| Key limitations | Small $n$; survey fatigue; halo effects; departure lounge bias | Self-selection; no face-to-face control | Bimodal distribution; selection bias; limited attribute coverage | Extreme selection bias; attribution ambiguity; negativity bias |
| Key ASQ papers | [17, 25] | [19, 56] | [7] | [7, 57] |

Note: NLP—naturallanguage processing; ACI—airports council international; ASQ—airport service quality; API—application programming interface.

Dalla Valle proposes a methodological contribution specifically aimed at addressing the sample bias problem of social media data by calibrating social media-derived satisfaction scores against contemporaneous ACI ASQ survey results [58], using airport-quarter as the unit of analysis. The calibration approach shows promise for extending the temporal and geographic coverage of ASQ measurement at a lower cost than traditional surveys, particularly for airports outside the ACI ASQ programme.

## 11 Airline Service Quality and Its Linkage with Airport Service Quality

The boundary between airline service quality and ASQ is conceptually clear—airlines control the aircraft cabin, check-in process, and boarding procedure; airports control the terminal, security, immigration, and landside access—yet in practice, passengers frequently conflate the two. This section reviews the evidence on how airline and ASQ interact and are perceived.

### 11.1 Attribution Ambiguity at the Airline—Airport Interface

Several service touchpoints straddle the airline–airport boundary. Check-in desks are airline-staffed but airport-located. Baggage delivery is an airline responsibility (carrier-to-carousel) but occurs at airport facilities. Lounge access is airline-controlled but physically within airport real estate. Boarding gates are airport infrastructure, but boarding is managed by airline ground staff. This creates systematic attribution ambiguity in passenger satisfaction surveys: when a passenger rates "check-in service quality", they may be evaluating the airport's physical infrastructure (counter layout, kiosk availability), the airline's staff performance, or both [1].

ASQ explicitly addresses this boundary, arguing for a reconceptualised ASQ framework that separates airport-controlled attributes (physical environment, landside access, security, wayfinding) from airline-influenced attributes (check-in efficiency, boarding, baggage) and provides a distinct measurement approach for each category [1].

### 11.2 Airport Service Quality Effects on Airline Brand Perceptions

Mainardes et al. [59] demonstrate that ASQ directly affects the corporate image of the airports themselves, and indirectly affects passengers' electronic word-of-mouth intentions—the likelihood of recommending or criticising the airport online. More importantly, several studies have found that **ASQ experiences "spill over" into passenger evaluations of connecting airlines**, particularly when the airport is strongly associated with a home carrier (hub-carrier relationships). Passengers at hub airports (e.g., Changi–Singapore Airlines; Incheon–Korean Air; Dubai–Emirates) tend to associate the airport experience more strongly with the airline's brand [21].

### 11.3 Studies Specifically Measuring Airport Quality from the Airline Perspective

The airlines folder contains a paper titled "Measuring airport quality from the airlines' viewpoint"—an important perspective that complements passenger-centric studies. Airlines evaluate airports on different criteria: slot availability, handling efficiency, baggage reconciliation time, gate proximity, ground handling quality, and aeronautical charges. The alignment (or misalignment) between airline-centric and passenger-centric airport performance measures is an underexplored area in the academic literature. Models study [29] extends the comparison to the airline service quality domain, providing a bridge between the two literatures, as mapped in Table 11.

**Table 11.** Airline service quality vs. ASQ—interface mapping

| Service Touchpoint | Primary Responsibility | Passenger Attribution | Research Evidence |
|---|---|---|---|
| Pre-flight online check-in | Airline | Airline | [1, 21] |
| Airport check-in counter | Airline (staff)+Airport (facility) | Mixed/often airline | [2, 17, 25] |
| Self-service check-in kiosks | Airline (system)+Airport (kiosk placement) | Mixed | [21, 38, 39] |
| Baggage drop | Airline (procedure)+Airport (belt/counter) | Mixed/often airline | [5, 25] |
| Security screening | Airport / Government | Usuallv airport | [6, 17, 31] |
| Departure lounge/gate area | Airport | Airport | [1, 21, 57] |
| Boarding process | Airline (procedure)+Airport (gate) | Airline | [25, 35] |
| Aircraft cabin experience | Airline | Airline | [21, 35] |
| Baggage claim delivery | Airline (ground handler)+Airport (carousel) | Mixed/often airline | [2, 5] |
| Lounge access and quality | Airline (access policy)+Airport (concession) | Airline | [1, 35] |
| Flight delays | Airline/ATC/Weather | Airline | [6, 7] |
| Landside ground transport | Airport + Operators | Airport | [2, 5, 27] |

Note: ATC—air traffic control; ASQ—-airport service quality.

## 12 Comparative Assessment: Academic Frameworks vs. Industry Indices

Several parallel systems for evaluating ASQ coexist, each with distinct methodologies, coverage, and intended audiences. This section compares the three most widely referenced systems and assesses their relative strengths and limitations for research and management purposes.

### 12.1 Airports Council International Airport Service Quality Programme

The ACI ASQ Programme, established in 2006, is the global benchmark for airport passenger satisfaction measurement. With participation from 340+ airports across 90 countries—collectively handling approximately 85% of world passenger traffic—it represents the most comprehensive standardised dataset available for airport performance comparison. The programme surveys approximately 650 passengers per quarter per airport (departing only) across 34 standardised attributes, using a 5-point scale (1 = Excellent to 5 = Poor, reversed for analysis). Results are published quarterly as the ACI ASQ Rankings and annually as regional benchmarks.

The 34 ACI ASQ attributes cover six service categories and are mapped to individual terminal processing and non-processing domains. AAI participates in the ACI ASQ programme, and quarterly parameter-wise scores are publicly available [26]. Analysis of this data reveals that Indian airports cluster in the middle tiers of global rankings, with GMr Group-operated Delhi and Mumbai performing best, while secondary and tertiary airports (operated by AAI directly) show consistent underperformance on check-in queue management, washroom cleanliness, and ground transportation access.

### 12.2 Skytrax World Airport Awards

The Skytrax World Airport Awards, derived from the Skytrax online passenger survey (typically 12–14 million surveys per annual cycle), rank airports globally across a broader set of criteria than ACI ASQ. While widely recognised in media and industry, Skytrax ratings face methodological challenges: lack of published sampling protocols, no quota sampling by passenger type, selection bias inherent to online self-selection, and commercial relationships between Skytrax and some ranked airports that create potential conflicts of interest. Alanazi et al. [7] provide the most rigorous academic assessment of Skytrax data, finding that while aggregate star ratings correlate reasonably with ACI ASQ scores, attribute-level patterns differ substantially due to the different passenger populations and measurement approaches.

### 12.3 J.D. Power Airport Satisfaction Study (North America)

J.D. Power's North America Airport Satisfaction Study covers large (≥30 million annual passengers), medium (10–30 million), and small (2.5–10 million) airport segments, surveying approximately 27,000 passengers annually across eight performance factors: terminal facilities, baggage claim, check-in/baggage check, security check, food, beverage and retail, airport accessibility, and the overall airport experience. The study is notable for segmenting results by passenger demographics and providing separate satisfaction indices for originating and connecting passengers. J.D. Power uses a 1,000-point scale that provides greater discrimination than the 5-point scale used by ACI ASQ.

### 12.4 Comparative Analysis

A comparative assessment of major ASQ measurement frameworks was carried out, covering the ACI ASQ Programme, Skytrax World Airport Awards, J.D. Power Airport Satisfaction Studies (North America), and prominent academic survey-based studies. The key findings from this assessment are summarised in Table 12 below, highlighting their respective methodologies, scope, strengths, limitations, and applicability for evaluating ASQ and passenger experience.

**Table 12.** Comparative assessment of major ASQ measurement indices

| Parameter | ACI ASQ Programme | Skytrax World Airport Awards | J.D. Power (North America) | Academic Survey Studies |
|---|---|---|---|---|
| Airports covered | 340+ globally | All major airports (self-nominated) | ∼35 North American airports | Single or few airports (most studies) |
| Annual surveys | ∼650/quarter/airport | 12–14 million (global, online) | ∼27,000 annually | 200–600 per study |
| Passenger type covered | Departing only | All (self-selected reviewers) | Originating + connecting | Mostly departing; some arriving |
| Attributes measured | 34 (standardised) | ∼46 (semistructured) | 8 performance factors | 8–74 (study-specific) |
| Rating scale | 5-point | 10-star + yes/no | 1000-point index | 5- or 7-point Likert |
| Sampling protocol | Quota sampling by flight; trained enumerators | Online self-selection; no quota | Online panel; structured quota by airport segment | Convenience/systematic; varies |
| Temporal frequency | Quarterly | Annual | Annual | One-time cross-section (majority) |
| Publication access | Paid subscription; summary rankings public | Public (website) | Public (press release); full report paid | Open/journal access |
| Longitudinal/ capability | Yes (time series since 2006) | Yes (since 1999) | Yes (annual) | Limited (cross-sectional majority) |
| Theoretical rigour | Moderate (industry-developed) | Low (no published methodology) | Moderate (published methodology) | High (peer-reviewed validation) |
| India coverage | Yes (AAI data publicly available) | Yes (some Indian airports) | No | Limited (few India-specific studies) |

Note: ACI—airportscouncil international; ASQ—airport service quality; AAI—airport authority of India.

**Key Finding:** Academic frameworks and industry indices measure broadly similar constructs but differ substantially in sampling rigour, attribute specificity, and theoretical grounding. The ACI ASQ Programme offers the best combination of standardisation, global coverage, and temporal depth, but its departure-only focus and 5-point scale are recognised limitations. Academic studies provide greater theoretical rigour, methodological diversity, and attribute-level depth, but their small sample sizes and single-airport focus limit generalisability. A hybrid approach—using ACI ASQ data as the baseline with supplementary academic survey instruments for deeper attribute analysis—represents the most robust strategy for comprehensive ASQ assessment, particularly in the Indian context.

Finally, emerging analytical approaches, including deep learning-based sentiment analysis, network analysis of airport systems, agent-based simulation of passenger flows, and natural language processing of multi-lingual passenger feedback, are expanding the methodological toolkit available to ASQ researchers. These approaches complement rather than replace traditional survey-based methods, offering scale, speed, and granularity advantages.

Resilience-oriented assessment has gained prominence following COVID-19, with researchers examining how airports maintained service quality during extreme demand fluctuations, staffing shortages, and evolving health protocols. The concept of "service quality resilience"—the ability to maintain acceptable service levels during disruptions—represents a novel extension of traditional ASQ frameworks.

Integrated data environments combining operational data (airport collaborative decision making systems), passenger tracking data (WiFi/Bluetooth), commercial transaction data, and survey data are enabling holistic, multi-source service quality assessment. This convergence addresses the longstanding limitation of single-source ASQ measurement.

The most recent period of ASQ research (2022–2025) has been characterised by several notable developments. Smart airport systems—integrating biometric processing, automated border control, self-service bag drops [38, 39], and contactless journey concepts—have emerged as a dominant research theme, driven by post-pandemic operational requirements and passenger expectations for minimal-touch travel experiences [57].

## 13 Research Gaps and Future Directions

To highlight the significance of the study, a comprehensive gap analysis was undertaken across the following key dimensions: landside service quality, arriving passenger experience, the specific context of Indian airports, domestic versus international airport comparisons, longitudinal/panel data analysis, integration of objective operational data, and inclusivity considerations, including services for persons with reduced mobility (PRM). These areas, summarised in Table 13, represent important research gaps that warrant further investigation to enhance the understanding and improvement of ASQ and passenger experience.

**Table 13.** Critical research gaps in domain-level Airport Service Quality literature

| Gap Area | Current State of Research | Recommended Direction | Relevance to India |
|---|---|---|---|
| Landside service quality | Fewer than 12% of papers address landside domains specifically; kerbside and ground transportation almost entirely absent | Dedicated landside ASQ index integrating objective (transport wait time, car park proximity) and subjective (satisfaction) data | Critical: Indian airport landside connectivity is a consistent pain point in ACI ASQ data |
| Arriving passenger experience | ∼15% of studies; baggage delivery most studied; arrivals lounge and egress almost absent | Dual-phase (departure + arrival) ASQ model with separate attribute weights for each phase | High: international arrivals at Indian airports face immigration, customs, and ground transport challenges |
| Indian airports specifically | 50 papers by Indian authors; majority cover foreign airports; Indian domestic terminals virtually unstudied | Indian airport-specific framework using AAI ACI ASQ data + primary surveys; Tier-2/3 airport studies | Critical: India is the world's third-largest aviation market |
| Domestic vs. international comparison | Most studies do not separate domestic and international passengers systematically | Comparative studies with separated domestic/international survey instruments | High: India's predominantly domestic market requires domestic-specific frameworks |
| Longitudinal/panel data | Almost all studies are cross-sectional snapshots | Panel surveys pre/post-major events (privatisation, terminal opening, COVID recovery) | High: Indian airport privatisation provides natural experimental variation |
| Objective data integration | ∼18% of studies use objective data; mostly DEA efficiency studies rather than passenger experience data | Sensor-integrated ASQ models using queue sensors, CCTV analytics, IoT data | Moderate: Smart City and DigiYatra initiatives create infrastructure for this |
| Inclusivity and PRM | Fewer than 3% of papers address disability-specific ASQ | PRM-specific service quality framework aligned with DGCA/IATA accessibility standards | High: The Rights of Persons with Disabilities Act, and 2016 mandates accessibility standards |

Note: ACI—Airports Council International; ASQ—Airport Service Quality; PRM—Persons with Reduced Mobility; DEA—Data Envelopment Analysis; AAI—Airport Authority of India; COVID—Coronavirus Disease; CCTV—Closed-Circuit Television; DGCA—Directorate General of Civil Aviation; IATA—International Air Transport Association.

Questionnaire-based studies, social media analyses, and industry benchmarking frameworks each involve distinct methodological constraints: survey-based studies suffer from response bias, recall decay, and departure-only sampling; social media platforms exhibit severe demographic skew and negativity bias; and industry indices lack methodological transparency and academic rigour. Cross-method triangulation—combining validated survey instruments with social media text mining and objective operational data—represents the most promising direction for overcoming these individual limitations. Conjoint-based approaches further offer structured alternatives for eliciting precise service quality attribute weights [60]. Table 14 summarises the principal limitations of current ASQ research approaches and identifies corresponding research directions.

**Table 14.** Comparative limitations of ASQ research approaches and future directions

| Research Approach | Principal Limitations | Recommended Future Directions |
|---|---|---|
| Questionnaire-based surveys (Intercept/Online) | Single-airport, cross-sectional designs limit generalisability; departure-lounge convenience sampling excludes arriving/transit passengers; common-method bias; low response rates (15–35%); cultural and language biases | Multi-airport longitudinal panel designs; stratified sampling across departure, arrival, and transit; mixed-method triangulation; adaptive instruments with demographic segmentation; standardised multilingual instruments |
| Social media analytics (Skytrax, Twitter/X, TripAdvisor) | Severe self-selection bias (extreme opinions overrepresented); demographic skew toward younger, English-speaking travellers; no control over attribute coverage; sentiment analysis accuracy limited by sarcasm/slang; platform policy changes | Calibration against validated survey benchmarks [31, 32]; demographic weighting and bias-correction; multi-platform fusion; domain-specific aspect-based sentiment analysis; real-time dashboard integration |
| Industry benchmarking (ACI ASQ, Skytrax, J.D. Power) | ACI ASQ is limited to departure only; Skytrax methodology is non-transparent with potential commercial bias; J.D. Power is restricted to North America; none integrates objective operational data; limited passenger profile disaggregation | Extend ACI ASQ to arrival and transit stages; develop a transparent, peer-reviewed global index; integrate objective KPIs alongside perceptions; terminal-level and segment disaggregation; open-access data sharing |

Note: ACI—Airports Council International; ASQ—Airport Service Quality; KPIs—key performance indicators

## 14 Summing Up

This paper has presented a comprehensive synthesis of the ASQ literature with a focus on service domain classification, attribute-level KPIs, passenger profile heterogeneity, survey methodology characteristics, social media data sources, airline—ASQ interface, and industry benchmarking indices. Several overarching conclusions emerge.

First, the processing/non-processing domain framework remains the most theoretically coherent and practically useful organisational structure for ASQ research and management. Processing domains are threshold-critical (must-be performance) while non-processing domains offer differentiation and competitive advantage (attractive attributes).

Second, passenger heterogeneity—by gender, age, travel purpose, nationality, class, and passenger type—is pervasive and significant. Generic ASQ frameworks that do not account for passenger segmentation will systematically misallocate resource priorities. The development of demographically segmented, context-specific ASQ indices represents a major opportunity for both research and practice.

Third, the convergence of social media platforms (Skytrax, Twitter) with traditional survey data is creating new opportunities for large-scale, longitudinal, and near-real-time ASQ monitoring. However, the severe selection biases of social media data require calibration against validated survey instruments before actionable conclusions can be drawn.

Fourth, the airline—ASQ interface remains poorly understood, with persistent attribution ambiguity at shared touchpoints. A reconceptualised, boundary-sensitive ASQ framework that explicitly delineates airport-controlled versus airline-controlled attributes would significantly improve the actionability of ASQ measurement for airport operators.

Fifth, the ACI ASQ Programme, while providing the most comprehensive global benchmark, has important limitations—particularly its departure-only focus, which excludes the arrival experience and landside domains. Academic research and industry practice would both benefit from extending the ACI ASQ framework to cover arriving passengers and landside service quality, with particular urgency for the growing Indian aviation market.

### 14.1 Implications for Airport Infrastructure Decision-Making

The service quality indicators synthesised in this review have direct implications for airport infrastructure planning and operational decision-making. Processing domain scores—particularly for check-in, security screening, and immigration—serve as leading indicators of terminal capacity constraints, informing decisions on facility expansion, technology deployment (e.g., automated kiosks, e-gates), and resource allocation during peak periods.

At the operational level, real-time ASQ monitoring—enabled by IoT sensors, social media sentiment tracking, and automated survey systems—supports dynamic resource allocation, queue management interventions, and service recovery protocols. At the strategic level, composite ASQ indices inform airport master planning, concession contract negotiations, regulatory compliance benchmarking (particularly for public-private partnership airports), and capital expenditure prioritisation across multi-terminal facilities.

Non-processing domain scores—ambient environment, wayfinding, retail, and food services—guide terminal design and renovation priorities, commercial space planning, and passenger flow management strategies. The stage-based aggregation framework (departure, transit, arrival) enables terminal managers to identify which journey phases require infrastructure investment versus operational process improvement.

### 14.2 Emerging Intelligent Technologies in Airport Infrastructure

Artificial intelligence and machine learning are being applied to predictive analytics for queue waiting times, automated sentiment analysis of passenger feedback, computer vision-based crowd density monitoring, and adaptive resource allocation systems. These approaches have shown promise at major hubs, enabling a shift from reactive to proactive service quality management.

The future of ASQ assessment is increasingly intertwined with emerging intelligent technologies that are transforming airport infrastructure systems. Digital twin technology enables the creation of virtual replicas of airport terminals, allowing operators to simulate passenger flows, test infrastructure modifications, and predict service quality impacts before physical implementation. Singapore Changi, Amsterdam Schiphol, and Dubai International Airport have begun deploying digital twins for terminal operations management.

### 14.3 Recent Trends in Airport Service Quality Research (2022–2025)

Resilience-oriented assessment frameworks—incorporating disruption scenarios (weather events, security incidents, pandemic protocols, system failures) into ASQ evaluation—are emerging as critical for airports operating in increasingly volatile environments. Adaptive management approaches that dynamically adjust service delivery based on real-time conditions are gaining traction.

Intelligent monitoring systems—combining IoT sensors, CCTV analytics, Bluetooth/WiFi tracking, and BMS data—provide continuous, objective measurement of service parameters (waiting times, ambient conditions, walking distances, dwell times) that complement subjective passenger perception surveys. Integrating these objective data streams with traditional ASQ survey data is a key frontier for future research.

### Author Contributions

Conceptualization, D.D. and S.G.; methodology, D.D.; software, D.D.; validation, D.D. and S.G.; formal analysis, D.D.; investigation, D.D.; resources, D.D.; data curation, D.D.; writing—original draft preparation, D.D.; writing—review and editing, S.G.; visualization, D.D.; supervision, S.G.; project administration, S.G. All authors have read and agreed to the published version of the manuscript.

### Data Availability

The data supporting the findings of this systematic literature review are available within the article. The complete list of 303 papers reviewed is available from the corresponding author upon reasonable request.

### Conflicts of Interest

The authors declare no conflicts of interest.

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